

Indoor Localization System with NLOS Mitigation Based on Self-Training

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Abstract—Location-awareness has become a fundamental requirement for multiple emerging applications with the rapid development of wireless technologies. The high-accuracy ranging enabled by ultra-wide bandwidth (UWB) signals is often deteriorated by clocks imperfections and non-line-of-sight (NLOS) propagation. Existing supervised learning methods for NLOS identification and mitigation are time-consuming, labor-intensive, and cost-inefficient due to the need for training data acquisition and label assignment. This paper presents an indoor localization system that enables NLOS mitigation based on self-training. The system provides a general information fusion framework that integrates map, inertial sensors, and UWB measurements, where the weak labels for UWB measurements are produced and iteratively refined by multi-sensory information fusion for self-training. In addition, the system utilizes the maximum likelihood ranging estimator that considers the impact of clock drift. The effectiveness of the proposed system is demonstrated via extensive experimentation in multiple real-world environments, e.g., the proposed methods reduce the NLOS ranging error by 80% and result in a 90th localization error percentile of 0.5 meters in a complex indoor environment.

Index Terms—Indoor localization, NLOS mitigation, self-training, weak supervision, information fusion.

1 INTRODUCTION

NETWORK LOCALIZATION has become the basis of multiple emerging applications, such as autonomous vehicles, smart cities, search-and-rescue operations, and environmental monitoring [1], [2], [3], [4], [5]. Global navigation satellite systems (GNSS) can meet the demand for outdoor navigation, but become unreliable in indoor and urban environments due to severely attenuated satellite signals [6], [7], [8]. Hence, numerous localization solutions have been developed in recent years based on other technologies, including radio frequency (RF) [9], [10], [11], inertial sensors [12], cameras [13], and ultrasonic waves [14].

Among existing RF techniques, ultra-wide bandwidth (UWB) possesses outstanding competitive advantages such as high transmission rate, strong anti-interference ability, low power consumption, and confidentiality [3]. Benefited from its wide bandwidth and high time-resolution characteristic, UWB radios can achieve centimeter-level ranging accuracy under line-of-sight (LOS) conditions [5]. Nevertheless, UWB ranging is affected in practice by unsynchronised clock between transceivers [15], [16] and non-line-of-sight

(NLOS) propagation [17], [18], [19]. In particular, NLOS conditions can lead to large ranging errors that significantly degrade the localization performance.

NLOS identification and mitigation are the key to reduce the impact of NLOS propagation on UWB-based localization systems [20], [21], [22]. In particular, machine learning approaches can provide significant performance improvements. Techniques based on support vector machines (SVMs) show satisfactory performance based on extracted channel statistics from the received channel impulse response (CIR) [18]. Other techniques have been shown to be effective such as those based on Gaussian processes regression (GPR) [23], variational autoencoder (VAE) and natural-parameter networks (NPN) [24]. Recently, the concept of soft ranging information (SRI) established a general framework for network localization, using a variety of machine learning techniques for dimensionality reduction and generative model estimation [4], [25].

Data acquisition and label assignment are critical limitations of NLOS identification and mitigation techniques based on supervised learning. Existing methods need to collect a large amount of training data at different locations and measure ground-truth for labeling. Acquiring abundant pre-annotated samples is time-consuming, labor-intensive, and cost-inefficient. In addition, environmental dynamics require data re-collection and re-labeling, resulting in even higher costs. Recently, new machine learning approaches have emerged to address these problems based on weak supervision [26], [27]. Such techniques utilize non-standard training data with incomplete, inexact, or inaccurate labels, aiming to reduce the dependence on accurate labels in practical applications. The self-training strategy is proposed to take the learner's predictions as weak labels, and to use these samples in the subsequent training procedure to enhance the learner [28].

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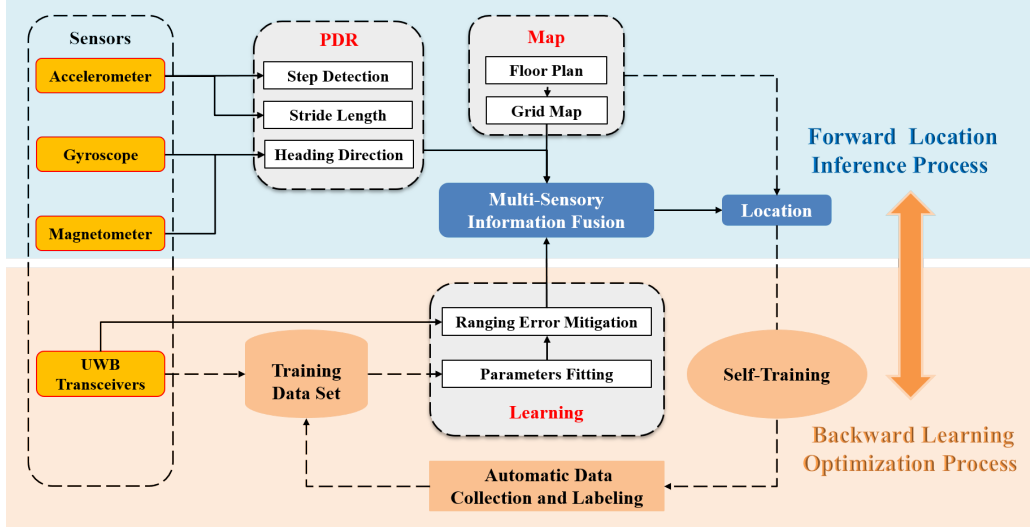


Fig. 1. The architecture of the proposed indoor localization system with NLOS mitigation based on self-training. The system operates by alternately executing the forward location inference process and the backward learning optimization process.

Weak supervision and self-training have been utilized to decrease the radio-map calibration cost of Wi-Fi-based localization systems. These systems usually take received signal strength indicator (RSSI) of Wi-Fi signals as fingerprint that is labeled with the corresponding location. In [29], a semi-supervised deep extreme learning machine (ELM) for fingerprint classification is proposed to adopt both labeled and unlabeled data by using Laplacian graph regularization. In [30], a semi-supervised deep reinforcement learning method is proposed to infer the classes of unlabeled RSSI data by incorporating a VAE. In [31], a semi-supervised co-localization method is proposed based on manifold learning and weighted k-nearest-neighbor (KNN) to match fingerprint. In [32], an online semi-supervised support vector regression (SVR) method is proposed to extend the graph-based semi-supervised learning to online learning and estimate unknown locations by fingerprint mapping. In [33], an unsupervised localization method is proposed by grouping unlabeled RSSI data and inertial navigation results to model the transition in Kalman filtering through fingerprint matching. In summary, most of such methods are based on semi-supervision that requires to use some labeled data and thus still need annotation effort. In addition, the existing methods obtain location information based on fingerprinting that requires to construct a costly database. Few works apply weak supervision or self-training on NLOS mitigation and UWB-based localization system. A semi-supervised learning approach for UWB ranging error mitigation has been recently proposed in [34], which leverages low-cost unlabeled measurements by self-training. However, such method relies on a certain portion of labeled samples to initialize the learner and requires training the additional learners for each unlabeled sample to improve the learner. Moreover, there is no performance evaluation of such method on the practical localization systems. The comparison of existing methods and the proposed method is summarized in Table 1.

In this paper, we propose an NLOS mitigation method based on self-training that is embedded in a multi-sensory information fusion localization system (see Fig. 1). Our

TABLE 1
Comparison of existing methods and the proposed method.

Method	Labeled data	Fingerprint	Information fusion	NLOS mitigation
[29] - [32]	✓	✓		
[33]		✓	✓	
[34]	✓			✓
This paper			✓	✓

system enables self-training by using automatic data collection and weak-labeling through multi-sensory information fusion. The requirement of labeled data is avoided, and the self-training procedure does not need to train additional learners. The proposed learning method is integrated into the localization system for evaluating the performance improvement. The main contributions are summarized as follows.

- We develop a weak-labeling method that automatically collects unlabeled data and obtains weak labels using map and inertial sensors.
- We improve the learner by iteratively refining weak labels through multi-sensory information fusion.
- We derive the maximum likelihood ranging estimator considering the influence of non-ideal clock drift.
- We conduct extensive real-world experiments that demonstrate the effectiveness of our method in terms of NLOS mitigation as well as the overall localization performance.

The paper is structured as follows. Section 2 describes the problem formulation and the localization system. We detail the proposed method for NLOS mitigation based on self-training in Section 3, and evaluate the system performance in Section 4. Finally, Section 5 concludes this paper.

2 SYSTEM OVERVIEW

In this section, we first present the problem formulation and the system architecture. Then we describe the main components of the proposed localization system, including map pre-processing, pedestrian dead reckoning (PDR) and UWB ranging. Finally, we will introduce the proposed self-training based NLOS mitigation method.

2.1 Problem formulation and system architecture

We consider two-dimensional localization of pedestrians through information fusion of map, inertial sensors and UWB measurements. The positional state of an agent at the k -th time is denoted by $\mathbf{s}_k = [\mathbf{p}_k^T, \theta_k]^T \in \mathbb{R}^3$ that is composed by the agent's location coordinates $\mathbf{p}_k \in \mathbb{R}^2$ and heading direction $\theta_k \in \mathbb{R}$. The UWB ranging measurements at k -th time are denoted by $\mathbf{z}_k$,¹ and the information from map and inertial sensors is used to model the change in the positional state over time.

Location inference is achieved based on the general probabilistic framework of recursive Bayesian inference [35], [36], [37]. The unknown conditional probability distributions of the state are estimated through two steps: prediction and update. In the prediction step, the probability distribution of current state conditioned on historical measurements is calculated using state transition function and conditional distributions of previous states. In the update step, the predicted distribution is updated by the likelihood of current observation to obtain the posterior conditional distribution. These two steps can be performed recursively, expressed as

$$p(\mathbf{s}_k | \mathbf{z}_{1:k-1}) = \int p(\mathbf{s}_{k-1} | \mathbf{z}_{1:k-1}) p(\mathbf{s}_k | \mathbf{s}_{k-1}) d\mathbf{s}_{k-1} \quad (1)$$

$$p(\mathbf{s}_k | \mathbf{z}_{1:k}) \propto p(\mathbf{s}_k | \mathbf{z}_{1:k-1}) p(\mathbf{z}_k | \mathbf{s}_k) \quad (2)$$

where the information from map and inertial sensors is used to calculate state transition function $p(\mathbf{s}_k | \mathbf{s}_{k-1})$, and the UWB ranging measurements are used to calculate the observation function $p(\mathbf{z}_k | \mathbf{s}_k)$, which becomes $p(\mathbf{z}_k | \mathbf{p}_k)$ since UWB ranging does not depend on the heading direction. The details of the state transition and the observation functions are given in the following subsections.

As illustrated in Fig. 1, the system architecture contains two processes: forward location inference process and backward learning optimization process. The former fuses multi-sensory information of map, PDR and UWB for location estimation. These three main components will be described in detail in Section 2.2 and 2.3. The latter introduces a feedback path from location estimates to model learning based on the proposed self-training method. This method is the main contribution of the paper and will be highlighted in Section 3. The entire localization system operates by alternately executing the two processes.

2.2 Map pre-processing and PDR

Map information in combination with PDR can filter out infeasible locations such as crossing buildings, walls, and

furniture. In order to incorporate map information efficiently, we adopt the scheme of the multi-grained grid-based filter proposed in [38], [39], which is used to develop a real-time localization system based on the PDR algorithm [40], [41].

We first perform map pre-processing to obtain discretized grid points. The process is as follows: (i) segments and layers information is extracted from the floor plan, (ii) grid map is obtained using uniform discretization, and (iii) connectivity among grid points is stored. We denote the map constraints as $M(\mathbf{p}^{j_p} \rightarrow \mathbf{p}^{i_p})$, which represents whether the grid point $\mathbf{p}^{i_p}$ can be reached from $\mathbf{p}^{j_p}$, where $\{\mathbf{p}^{i_p}\}_{i_p=1}^{N_p}$ are the N_p uniformly discretized grid points of the locations on the map. If the two grid points are blocked by obstacles, or the distance between them exceeds a threshold $d_{\max}$, $M(\mathbf{p}^{j_p} \rightarrow \mathbf{p}^{i_p}) = 0$; otherwise $M(\mathbf{p}^{j_p} \rightarrow \mathbf{p}^{i_p}) = 1$.

The inertial sensor measurements are incorporated into the state transition by the PDR algorithm, which involves step detection, stride length estimation, and heading direction estimation. We detect steps by identifying adjacent peaks and valleys of the filtered acceleration data, then use frequency model [42] for stride length estimation and complementary filter [43] for heading direction estimation.

The multi-grained grid-based filter discretizes the state space and approximates the conditional distribution by the weighted sum [39]

$$p(\mathbf{p}_k, \theta_k | \mathbf{z}_{1:k}) \approx \sum_{(i_p, i_a)} w_{k|k}^{(i_p, i_a)} \delta(\mathbf{p}_k - \mathbf{p}^{i_p}) \delta(\theta_k - \theta^{i_a}) \quad (3)$$

where $\{\theta^{i_a}\}_{i_a=1}^{N_a}$ are the N_a uniformly discretized angles of the heading direction, $w_{k|k}^{(i_p, i_a)}$ represents the probability of positional state equal to $(\mathbf{p}^{i_p}, \theta^{i_a})$ at k -th time, and $\delta(\cdot)$ is the delta function. Then, the recursive process in (1) and (2) can be transformed into determining the prior and posterior weights

$$w_{k|k-1}^{(i_p, i_a)} = \sum_{(j_p, j_a)} w_{k-1|k-1}^{(j_p, j_a)} M(\mathbf{p}^{j_p} \rightarrow \mathbf{p}^{i_p}) \cdot p(\mathbf{p}_k^{i_p}, \theta_k^{i_a} | \mathbf{p}_{k-1}^{j_p}, \theta_{k-1}^{j_a}, \mathbf{m}_k) \quad (4)$$

$$w_{k|k}^{(i_p, i_a)} \propto w_{k|k-1}^{(i_p, i_a)} \cdot p(\mathbf{z}_k | \mathbf{p}_k^{i_p}) \quad (5)$$

where $\mathbf{m}_k$ includes the stride length l_k and the heading direction change $\Delta\theta_k$ of the k -th step estimated by the PDR algorithm.

Finally, current location estimates can be expressed as the weighted sum of coarse grid points

$$\hat{\mathbf{p}}_{k|k} = \frac{\sum_{(i_p, i_a)} w_{k|k}^{(i_p, i_a)} \mathbf{p}^{i_p}}{\sum_{(i_p, i_a)} w_{k|k}^{(i_p, i_a)}}. \quad (6)$$

2.3 UWB ranging

UWB transceivers first apply the ranging estimator to obtain the distance between the user and base station. In the subsection we present the maximum likelihood ranging estimator in the presence of clock drift.

For LOS scenarios, each ranging estimate of UWB measurements $\mathbf{z}_k$ at k -th time can be obtained by multiplying the time-of-flight (ToF) estimate $\hat{T}_f$ and signal propagation

1. In our implementation, multiple UWB measurements are obtained between consecutive steps of PDR due to higher ranging frequency. To reduce state update frequency, we use the average value of ranging estimates between two steps as the observation $\mathbf{z}_k$.

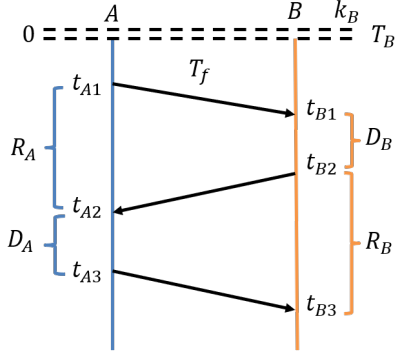


Fig. 2. Two-way ranging model for ToF estimate.

speed. Consider the two-way ranging model for ToF estimate between user A and base station B (see Fig. 2). The transmitting/receiving times for ground truth and measurement are represented by $t_{X,i}$ and $\hat{t}_{X,i}$ respectively, where $X \in \{A, B\}$ and $i \in \{1, 2, 3\}$. The measured time is affected by the clock drift, modelled as

$$\hat{t}_{X,i} = k_X t_{X,i} + T_X + n_{X,i} \quad (7)$$

where T_X is the clock offset, $k_X = 1 + e_X$ in which e_X denotes the deviation from real frequency, and the noise $n_{X,i}$ is assumed to follow a Gaussian distribution with zero mean and variance $\sigma_{X,i}^2$.

Without loss of generality, suppose A be the standard clock with initial time $T_A = 0$ and $k_A = 1$, whereas T_B and k_B are the clock offset and relative frequency of B with regard to A, respectively. Then, the unknown parameters can be represented by $\Theta = \{t_{A1}, t_{A2}, t_{A3}, T_f, T_B, k_B\}$, and all the six time measurements can be represented by $\mathbf{z} = \{\hat{t}_{A1}, \hat{t}_{A2}, \hat{t}_{A3}, \hat{t}_{B1}, \hat{t}_{B2}, \hat{t}_{B3}\}$, given by

$$\begin{aligned} \hat{t}_{Ai} &= t_{Ai} + n_{Ai}, \quad i = 1, 2, 3 \\ \hat{t}_{Bi} &= \begin{cases} T_B + k_B(t_{Ai} + T_f) + n_{Bi}, & i = 1, 3 \\ T_B + k_B(t_{Ai} - T_f) + n_{Bi}, & i = 2 \end{cases} \end{aligned} \quad (8)$$

Based on the measurements, we can obtain the maximum likelihood estimators of the ToF and clock drift as

$$\hat{T}_f = \frac{\alpha_3(\beta_1 - \beta_2) + \alpha_1(\beta_3 - \beta_2)}{2(\alpha_1 + \alpha_3)\hat{k}_B} \quad (9)$$

$$\hat{k}_B = \frac{\hat{t}_{B3} - \hat{t}_{B1}}{\hat{t}_{A3} - \hat{t}_{A1}} \quad (10)$$

where $\alpha_i = \sigma_{Ai}^2 \hat{k}_B^2 + \sigma_{Bi}^2$ and $\beta_i = \hat{t}_{Bi} - \hat{k}_B \hat{t}_{Ai}$ for $i \in \{1, 2, 3\}$.

Remark. As a special case when the standard deviations of transmitter/receiver noises are the same, i.e., $\sigma_{A1} = \sigma_{A3} = \sigma_{B2}$ and $\sigma_{A2} = \sigma_{B1} = \sigma_{B3}$, the ToF estimator can be simplified as

$$\begin{aligned} \hat{T}_f &= \frac{1}{4} [(\hat{t}_{A2} - \hat{t}_{A1}) - (\hat{t}_{A3} - \hat{t}_{A2})] \\ &\quad + \frac{1}{4\hat{k}_B} [(\hat{t}_{B3} - \hat{t}_{B2}) - (\hat{t}_{B2} - \hat{t}_{B1})], \end{aligned} \quad (11)$$

which can be shown equivalent to the classic double-sided two-way ranging method proposed in [44].

In NLOS propagation environments, the ToF estimator (9) will yield a positive ranging bias due to signal reflection, refraction or penetration of obstacles. For example, experiments show that the UWB ranging error in LOS scenarios are normally around 30cm to 50cm, while the NLOS ranging biases are larger than 1m for 20% times even in a simple indoor office, and can exceed 5m for 60% times at the corner of a corridor (see Fig. 7). These erroneous measurements can highly degrade the localization performance if they are not properly mitigated.

Multiple techniques have been proposed to fuse PDR, UWB and map information including [45], [46], [47]. Such methods either do not directly address the NLOS mitigation problem or utilize simplistic approaches based on line segment matching. In this paper, we propose a self-training method to solve the NLOS mitigation problem, which can automatically accomplish data collection, labeling, and learning through multi-sensory information fusion instead of a pre-deployed site survey. We will describe the proposed method in detail in Section 3.

3 NLOS MITIGATION BASED ON SELF-TRAINING

We propose a simple and effective method for NLOS mitigation based on self-training. Such method is the basis of the backward learning optimization process of our system as shown in Fig. 1. The techniques presented can tackle the difficulties of traditional supervised learning methods in data acquisition and label assignment.

The proposed self-training method is composed of three steps: (i) automatic data collection and weak-labeling; (ii) model learning; and (iii) database and model update. In the first step, our system automatically collects unlabeled UWB data indexed by time and produces weak labels according to location estimates from map and inertial sensors. In the second step, machine learning algorithms are applied to obtain NLOS models trained with the database consisting of collected data and weak labels. These NLOS models include NLOS identification model and NLOS mitigation model. The former is trained with weak condition labels to obtain a classification algorithm for the purpose of estimating the LOS/NLOS channel condition. The latter is trained with weak distance labels to obtain a regression algorithm for the purpose of estimating the distance between the user and base station. In the third step, the database and models are updated iteratively through multi-sensory information fusion. In the following subsections, we describe the three steps in detail.

3.1 Automatic data collection and weak labeling

In order to realize NLOS ranging error mitigation with machine learning techniques, a large amount of data needs to be collected and labeled to construct training database.

Automatic data collection. Our system records the received UWB measurements in real time during the positioning process. These measurements include: (i) received timestamp t ; (ii) sampled CIR waveform $\mathbf{r}_t$ (UWB data); (iii) ranging estimate $\hat{d}_t$ calculated by the maximum likelihood estimator in Section 2.3; and (iv) ID of corresponding base station b_t . Abundant unlabeled data can be easily obtained during continuous walking behavior.

Automatic weak-labeling. Our system produces weak labels for the collected UWB data by using the map, current location estimates, and corresponding base station location. The system first performs timestamp matching on the collected UWB data and the location estimates at each step $\hat{\mathbf{p}}_k$ calculated by (6). Afterwards, the system annotates the condition label and the distance label for the purpose of NLOS identification and mitigation, respectively. The condition label is assigned as $\tilde{l} = +1$ for LOS condition if the line $\hat{\mathbf{p}}_k \longleftrightarrow \mathbf{p}_{b_k}$ is not intersected by any obstacle on the map, where $\mathbf{p}_{b_k}$ is the location of the base station at the k -th time step with ID b_k ; otherwise, it is assigned as $\tilde{l} = -1$ for NLOS condition. Similarly, the distance label is assigned as $\tilde{d} = \|\hat{\mathbf{p}}_k - \mathbf{p}_{b_k}\|$. Here we use the 2D map to have an estimate of the channel condition and distance between the user and corresponding base station. The estimation may be inaccurate sometimes due to different heights of obstacles, unknown occlusion of pedestrians, and error of positioning results. Therefore, the labels obtained may be inaccurate and referred to as “weak labels” in the paper.

We design two labeling schemes on the basis of above weak labels: the interpolation labeling scheme and the probability labeling scheme. Above weak labels can only label the aligned UWB data at each step. In practice, the UWB ranging frequency is usually higher than the PDR positioning frequency to bring out abandoned intermediate UWB data between $\hat{\mathbf{p}}_{k-1}$ and $\hat{\mathbf{p}}_k$. To deal with this problem, the interpolation labeling scheme performs linear interpolation between each two steps:

$$\tilde{\mathbf{p}}_i = \hat{\mathbf{p}}_{k-1} + \frac{\hat{\mathbf{p}}_k - \hat{\mathbf{p}}_{k-1}}{t_k - t_{k-1}}(t_i - t_{k-1}) \quad (12)$$

where $\tilde{\mathbf{p}}_i$ is a virtual position at time t_i (decided by the ranging frequency). These virtual positions are also used to produce weak labels.

The interpolation labeling scheme can help expand data size, but ignore probabilistic information when only using the expectation of coarse grid points. In contrast, our alternative probability labeling scheme can not only expand data sets, but also make full use of the weight distribution information over different grid points. More precisely, we arrange the grids of the k -th step in descending order of location beliefs $\sum_{i_a} w_{k|k}^{(i_p, i_a)}$, then select C candidate positions from the top, and obtain the relative probability $\{\text{prob}_1, \text{prob}_2, \dots, \text{prob}_C\}$ after normalization. The current matched data sample is copied $E \times \text{prob}_i$ times with labels calculated based on the corresponding candidate position, where E is the chosen expansion factor. Take $C = 2$ and $E = 10$ for example, after data matching at each step k , two samples are formed with weak labels calculated by candidate positions, denoted as $(\text{prob}_1 = 0.7, \tilde{d}_1 = 10, \tilde{l}_1 = -1)$ and $(\text{prob}_2 = 0.3, \tilde{d}_2 = 12, \tilde{l}_2 = -1)$. Then the former sample is copied 7 times and the latter one is copied 3 times, preserving the probability ratio of 7:3 in the training set.

In summary, we match the collected UWB data with location estimates utilizing aligned timestamps, and then produce weak labels using map information. The process of data collection and labeling is done automatically in the background without user intervention.

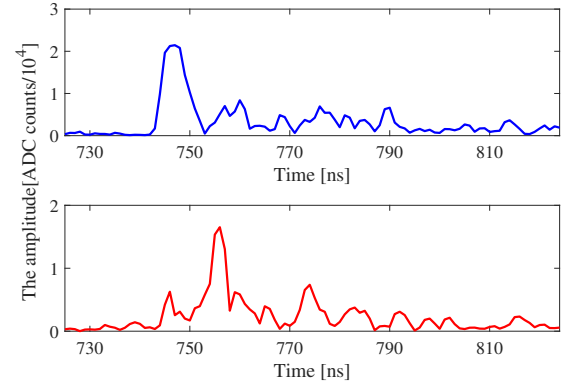


Fig. 3. Comparison of channel impulse response of real data collected under LOS (upper) and NLOS (lower) scenarios.

3.2 Model learning with self-training

Given the training database, various machine learning algorithms can be applied to estimate the mapping between input features and output labels. We first describe the features used to distinguish LOS and NLOS conditions, and then present a brief introduction to classification and regression.

Feature selection. Due to reflections and obstructions, the received UWB waveform presents different characteristics under LOS and NLOS conditions. Fig. 3 compares examples of LOS and NLOS channel impulse responses obtained in an indoor scenario. Generally, NLOS signals exhibit smaller energy, longer rise time, and larger temporal dispersion dominated by multipath components. Thus, several physical features can be extracted from the collected UWB data as input features to reduce dimension. Previous studies [18], [23], [25] have demonstrated seven useful extracted features, including: energy ξ_r , maximum amplitude $r_{\max}$, rise time t_{rise} , mean excess delay τ_{MED} , RMS delay spread τ_{RMS} , kurtosis κ , and ranging estimate $\hat{d}$. In this paper, we propose the ratio between the first path amplitude r_{FP} and the standard deviation of channel impulse response estimate noise σ_{CIRN} as another feature that assesses the quality of the received signal.² Therefore, a poor quality indicates a low first path intensity close to the noise level, which is likely to be obtained with NLOS condition. We provide the analytical expression of each feature in Table 2.

Classification. The training data set for classification is $D = \{(\mathbf{x}_n, \tilde{l}_n)\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbb{R}^d$ denotes the input features extracted from the received waveform $\mathbf{r}_n$, and $\tilde{l}_n \in \{+1, -1\}$ denotes the weak condition label assigned to $\mathbf{x}_n$. Multiple classification algorithms (see e.g., [49], [50], [51], [52]) can be used to train the NLOS identification model as described below.

SVM is one of the most widely used classification techniques based on the Vapnik–Chervonenkis theory and structural risk minimization principle [49], [50]. The goal of SVM is to find the maximum-margin hyperplane to maximize the separation between positive and negative training samples.

2. The term r_{FP} and σ_{CIRN} are given through the LDE algorithm’s analysis of the accumulator data and reported in DW1000 single chip wireless transceiver [48].

TABLE 2
Extracted features.

Feature	Formula
Energy	$\xi_r = \int_T r(t) ^2 dt$
Maximum amplitude	$r_{\max} = \max_t r(t) $
Rise time ^a	$t_{\text{rise}} = t_H - t_L$
Mean excess delay ^b	$\tau_{\text{MED}} = \int_T t \psi(t) dt$
RMS delay spread ^b	$\tau_{\text{RMS}} = \int_T (t - \tau_{\text{MED}})^2 \psi(t) dt$
Kurtosis ^c	$\kappa = \frac{1}{\sigma_{ r }^4 T} \int_T [r(t) - \mu_{ r }]^4 dt$
Ranging estimate	$\hat{d}$
Quality	$r_{\text{quality}} = \frac{r_{\text{FP}}}{\sigma_{\text{CIRN}}}$

^a $t_L = \min \{t : |r(t)| \geq \alpha \sigma_n\}$, $t_H = \min \{t : |r(t)| \geq \beta r_{\max}\}$, where σ_n is the standard deviation of the thermal noise. The values of $\alpha > 0$ and $0 < \beta \leq 1$ are chosen empirically in order to capture the rise time; in our case, we used $\alpha = 6$ and $\beta = 0.6$.

^b $\psi(t) = \frac{|r(t)|^2}{\xi_r}$.

^c $\mu_{|r|} = \frac{1}{T} \int_T |r(t)| dt$, $\sigma_{|r|}^2 = \frac{1}{T} \int_T [|r(t)| - \mu_{|r|}]^2 dt$.

This hyperplane is expressed as

$$y(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b \quad (13)$$

where $\phi(\cdot)$ is a feature map that describes a non-linear transformation.

After Lagrangian dual transformation and optimization, the final SVM decision model can be expressed as

$$t(\mathbf{x}) = \text{sign} \left(\sum_{n=1}^N \alpha_n^* \tilde{l}_n k(\mathbf{x}, \mathbf{x}_n) + b^* \right) \quad (14)$$

where $k(\mathbf{x}, \mathbf{x}_n)$ is the value for points $\mathbf{x}$ and $\mathbf{x}_n$ of the kernel function induced by the feature map $\phi(\cdot)$. According to KKT conditions, only the so-called support vectors on/within the margin contribute to the decision, and thus SVM possesses both robustness and sparsity.

Another widely used and rapidly developed classification techniques are based on artificial neural networks (ANNs) [52]. The multilayer perceptron (MLP) estimates the probability distribution $p(\mathbf{x}; \mathbf{w})$ through multi-layer linear connections and non-linear activation functions. The parameters are generally fitted through minimizing the cross-entropy loss function.

Regression. The training data set for regression is $D = \{(\mathbf{x}_n, d_n)\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbb{R}^d$ denotes the extracted features, and $d_n \in \mathbb{R}$ denotes the weak distance label. Multiple regression algorithms can be used to train the NLOS mitigation model as described below.

SVM can be further extended to regression task and optimized in a similar way to obtain the final regressor

$$t(\mathbf{x}) = \sum_{n=1}^N (\alpha_n^* - \hat{\alpha}_n^*) \tilde{d}_n k(\mathbf{x}, \mathbf{x}_n) + b^* \quad (15)$$

Algorithm 1: Indoor localization system with NLOS mitigation based on self-training

Input: PDR measurements $\mathbf{m}$, floor plan map, UWB measurements $\mathbf{z}$, iteration number I

Output: localization results $\hat{\mathbf{p}}$, UWB models

```

1 Map pre-processing;
2 Initialization;
3 for  $i = 0 : I - 1$  do
4   for  $k = 1 : K - 1$  do
5     Grid weights predict:
6     Predict weights according to (4);
7     if  $i == 0$  then
8        $w_{k|k} = w_{k|k-1}$ ;
9     else
10      Grid weights update:
11      Identify and mitigate NLOS data
12      according to models of  $i - 1$ ;
13      Calculate observation likelihood
14      according to (16);
15      Update weights according to (5);
16    end
17  end
18 Calculate estimated location according to (6);
19 Data alignment and labeling;
20 Model learning;
21 return  $\hat{\mathbf{p}}$  and models;
22 end

```

which only relates to partial support vectors with $\alpha_n^* \neq 0$ or $\hat{\alpha}_n^* \neq 0$, and thus is memory-efficient and fast. MLP can also be used for regression task through minimizing the quadratic loss function.

In summary, we extract several discriminative features from the collected UWB data, and then train the NLOS models based on machine learning algorithms. Notice that our method provides a learning framework, and it can adopt different classification and regression models flexibly.

3.3 Database and model update based on self-training

Given trained models, the NLOS effects can be mitigated to facilitate weak labels refinement through multi-sensory information fusion. We first introduce the designed observation likelihood function, and then describe the database and model update procedure.

Observation likelihood. The new collected UWB data is first identified whether coming from LOS or NLOS propagation. We design two NLOS identification schemes to distinguish channel conditions: the classifier-based and the map-based schemes. The former applies the classifier obtained in Section 3.2, while the latter checks whether there are obstacles between the current grid point and the corresponding base station on the map.

For the data identified as LOS, the system directly uses original high-precision ranging estimate $\hat{d}$ to calculate observation probability. Whereas, for the data identified as NLOS, the regressor obtained in Section 3.2 can provide

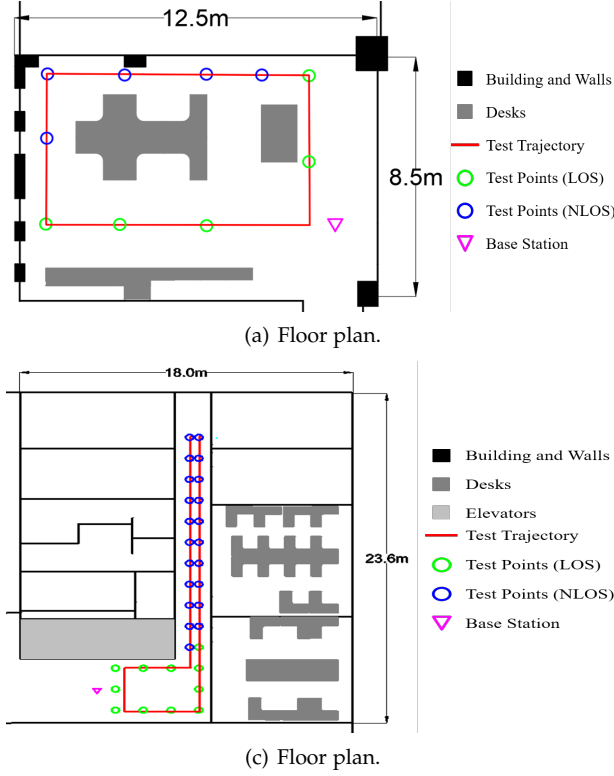


Fig. 4. Experimental sites. Test scene A is an indoor office shown in (a) and (b). Test scene B is a corridor corner shown in (c) and (d).

more reliable distance $\hat{d}^m$. The observation model in (5) is given by³

$$p(\mathbf{z}_k | \mathbf{p}_k^{i_p}) = \prod_{\hat{d}_{b_k} \in \text{LOS}} p_{\text{LOS}}(\hat{d}_{b_k}; \hat{d}_{b_k}^{i_p}, \sigma_{\text{LOS}}^2) \cdot \prod_{\hat{d}_{b_k} \in \text{NLOS}} p_{\text{NLOS}}(\hat{d}_{b_k}^m; \hat{d}_{b_k}^{i_p}, \sigma_{\text{NLOS}}^2) \quad (16)$$

where $\hat{d}_{b_k}$ and $\hat{d}_{b_k}^m$ represent the distance estimate and the mitigated distance between the user and the corresponding base station with ID b_k . They are assumed to follow a Gaussian distribution with the same mean value $\hat{d}_{b_k}^{i_p} = \|\mathbf{p}_k^{i_p} - \mathbf{p}_{b_k}\|$.

Database and model update. The UWB measurements are incorporated into the system for multi-sensory information fusion using the designed observation likelihood function. By substituting (16) into (5) and (6), we can obtain the updated weight distributions and the revised location estimates. The system then uses these revised location estimates to produce new weak labels. The steps described in Section 3.1-3.3 are executed iteratively based on the self-training strategy to update database and model. We demonstrate that this strategy can improve the quality of weak labels and is the key to high accuracy, low cost, and strong robustness of our system.

Finally, the overall proposed indoor localization system with NLOS mitigation based on self-training is summarized

3. The filtering algorithm requires that the maximum value of $p(\mathbf{z}_k | \mathbf{p}_k^{i_p})$ should not be smaller than a threshold, in which case UWB measurements are not used due to inconsistency with PDR results.

in Algorithm 1.⁴ In the proposed algorithm, an iteration refers to a model update period during the training process, which includes a complete procedure of localization, labelling and training. Such iteration is carried out to re-train the NLOS mitigation models when N_t trajectories or N_s UWB measurements are collected. As the number of iteration increases, the system tends to reach stability so that we can stop the iterations to reduce the computation complexity. In practice, the stopping criteria can be determined when the norm of the positioning difference between last iteration and current iteration does not exceed a certain threshold.

4 EVALUATION

In this section, we present the experimental results that assess the performance of the proposed system in terms of NLOS mitigation and indoor localization. We also discuss further generalization of our system to online learning and multi-base station localization.

In Sections 4.3-4.5 we show the results corresponding with a batch mode in which each iteration estimates all the trajectories, while in Section 4.6 we show the results corresponding with an online mode in which each iteration estimates one trajectory. When evaluating the localization performance, the baseline that we use for comparison is the multi-grained grid-based filter proposed in [39] as described in Section 2.2. We compare the localization performance against the baseline method using only map based PDR

4. The initial probability distribution of location can be obtained by using prior location information or other measurements such as Wi-Fi RSSI [39].

information, which is referred to as “Only PDR” in the following sections.

4.1 Experimental settings

We conduct real-world experiments in different indoor environments to assess the applicability of our system. *Test scene A* is a common indoor office with a dimension of $12.5\text{ m} \times 8.5\text{ m}$, as shown in Figs. 4a and 4b. The obstacles causing NLOS propagation mainly include desks and chairs. The user walks continuously along the test trajectory holding a smartphone and an UWB tag, with a total length of 728m. *Test scene B* is a corner of a corridor with a dimension of $18.0\text{ m} \times 23.6\text{ m}$, as shown in Figs. 4c and 4d. This scene contains multiple walls and metal occlusions with complex signal reflections and refractions. In different time periods within a month, 12 volunteers with different genders, heights, and weights collected 43 trajectories with a total length of 2021m. The test trajectory, test points, and the location of the base station are all marked in the figure. For the grid-based filtering, we utilize a grid with quantization interval of 0.5m. Such quantization offers an adequate compromise between accuracy and complexity. Larger quantization intervals can lead to a poor performance between doors and smaller intervals result in similar localization errors (see [39] for more details regarding such quantization).

Our UWB devices are based on the IEEE802.15.4-2011 compliant IR-UWB wireless transceiver DW1000, which implements the two-way ranging method described in Section 2.3 and is capable of sampling the CIR waveform at Nyquist rate. The center frequency, bandwidth and data transmission rate of the UWB radio are set to 3494.4 MHz, 499.2 MHz, and 6.8 Mbps, respectively. Each base station is installed on top of a tripod at the same height as the tags. In *Test scene A*, four antennas are used to receive signals concurrently, and a single antenna is used in *Test scene B*. We developed an Android APP on the smartphone to collect IMU readings and UWB measurements. When starting to record the inertial sensor data, it automatically subscribes to UWB measurements published by the server using MQTT protocol to realize time alignment.

4.2 Database

A database was created for NLOS identification and mitigation together with the aforementioned measurement apparatus and arrangements. The database is composed by received waveforms, extracted features, and weak labels obtained during positioning. When obtaining weak labels, the timestamp matching range is set to $\pm 0.1\text{s}$ considering that the ranging frequency is 10Hz. For interpolation labeling scheme, the linear interpolation time interval is set to 0.1s. For probability labeling scheme, the number of candidate positions is set to $C = 3$, and the expansion factor is set to $E = 10$. The size and labels of training data sets change dynamically during the self-training process.

We evaluate the performance of the models obtained by our system on the test points uniformly distributed in the positioning area, as shown in Fig. 4. At each point, a set of UWB data is recorded for 1 minute and labeled based on the real coordinates. For *Test scene A*, 5 points are under

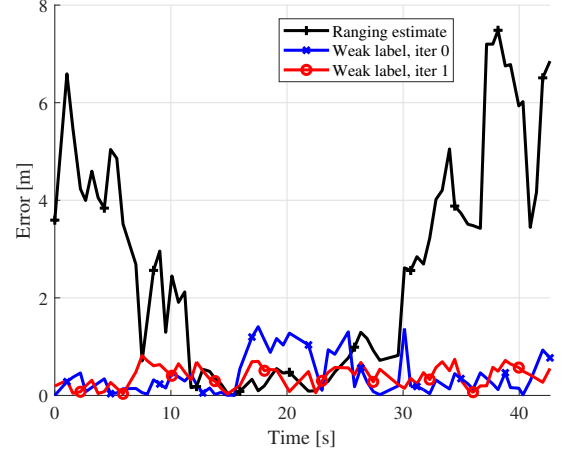


Fig. 5. Ranging errors and weak labels in the first two iterations.

LOS condition and 5 points are under NLOS condition, contributing to a total of 33753 test samples. The ranging errors have the CDF shown in Fig. 7a, and the root mean squared error (RMSE) of NLOS measurements is 0.91m. For *Test scene B*, 11 points are under LOS condition and 21 points are under NLOS condition, contributing to a total of 20397 test samples. The ranging errors have the CDF shown in Fig. 7b, and the root mean squared error (RMSE) of NLOS measurements is 5.27m.

4.3 Performance quantification

We first test and analyze the performance of each part of the proposed algorithm.

Automatic weak-labeling. We demonstrate the effectiveness of our automatic weak-labeling procedure and the label quality improvement achieved by self-training. Fig. 5 shows the error (absolute difference with the actual distances) of ranging estimates and weak labels in the first two iterations of one trajectory in *Test scene B*. In the first iteration (iter 0), the weak labels are obtained by only PDR positioning results to train machine learning models. In the second iteration (iter 1), the weak labels are obtained by “PDR + UWB” data fusion positioning results with NLOS mitigation using the models of iter 0. The user passes through the “NLOS $\rightarrow$ LOS $\rightarrow$ NLOS” area over time. We observe that ranging estimates show large positive biases under NLOS conditions, while the weak labels obtained by our system are much closer to the actual distances. In addition, the weak labels around 20s are significantly more accurate after one iteration. The RMSE of weak labels is reduced from 0.56m to 0.44m, showing an improvement in label quality that results in more accurate models.

Features selection. We discuss the impact of different feature combinations on model training. Table 3 shows the NLOS identification performance using SVM classification models trained with different feature sets in *Test scene A*, where each set $\mathcal{F}_i^j$ represents the best set of size i for the purpose of NLOS identification. Therefore, the ranking of features can be concluded as: $r_{\max} > \hat{d} > t_{\text{rise}} > \xi_r > \tau_{\text{RMS}} > \kappa > r_{\text{quality}} > \tau_{\text{MED}}$. Considering all feature set

TABLE 3

NLOS identification performance under different feature combinations. The set $\mathcal{F}_I^i$ represents the set of i features with the highest accuracy.

Set	Features	Precision [%]	Recall [%]	Accuracy [%]
$\mathcal{F}_I^1$	$r_{\max}$	99.34	81.98	91.06
$\mathcal{F}_I^2$	$\hat{d}, r_{\max}$	98.64	82.80	91.17
$\mathcal{F}_I^3$	$\hat{d}, t_{\text{rise}}, r_{\max}$	98.99	83.74	91.77
$\mathcal{F}_I^4$	$\hat{d}, \xi_r, t_{\text{rise}}, r_{\max}$	98.25	85.46	92.27
$\mathcal{F}_I^5$	$\hat{d}, \xi_r, \tau_{\text{RMS}}, t_{\text{rise}}, r_{\max}$	98.54	85.81	92.56
$\mathcal{F}_I^6$	$\hat{d}, \xi_r, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\max}$	98.32	86.10	92.60
$\mathcal{F}_I^7$	$\hat{d}, \xi_r, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\max}, r_{\text{quality}}$	97.44	83.92	91.20
$\mathcal{F}_I^8$	$\hat{d}, \xi_r, \tau_{\text{MED}}, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\max}, r_{\text{quality}}$	97.41	83.49	90.99

TABLE 4

NLOS mitigation performance under different feature combinations. The set $\mathcal{F}_M^i$ represents the set of i features with the lowest RMSE.

Set	Features	RMSE [m]	NLOS-RMSE [m]	R^2 [%]
$\mathcal{F}_M^1$	$\hat{d}$	1.28	1.06	0.75
$\mathcal{F}_M^2$	$\hat{d}, r_{\text{quality}}$	1.27	1.05	0.75
$\mathcal{F}_M^3$	$\hat{d}, t_{\text{rise}}, r_{\text{quality}}$	1.28	1.03	0.76
$\mathcal{F}_M^4$	$\hat{d}, \xi_r, t_{\text{rise}}, r_{\text{quality}}$	1.27	1.08	0.76
$\mathcal{F}_M^5$	$\hat{d}, \xi_r, \kappa, t_{\text{rise}}, r_{\text{quality}}$	1.26	1.08	0.76
$\mathcal{F}_M^6$	$\hat{d}, \xi_r, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\text{quality}}$	1.27	1.07	0.76
$\mathcal{F}_M^7$	$\hat{d}, \xi_r, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\max}, r_{\text{quality}}$	1.27	1.08	0.75
$\mathcal{F}_M^8$	$\hat{d}, \xi_r, \tau_{\text{MED}}, \tau_{\text{RMS}}, \kappa, t_{\text{rise}}, r_{\max}, r_{\text{quality}}$	1.28	1.09	0.75

sizes, we select $\mathcal{F}_I^6$ as the best combination for classification and employ it in the subsequent experiments.

Similarly, Table 4 shows the NLOS mitigation performance using SVM regression models trained with different features' combinations, where each set $\mathcal{F}_M^i$ represents the best set of size i for the purpose of NLOS mitigation. The coefficient of determination R^2 measures how well the regression predictions approximate the real data points and is expected to be close to 1. Therefore, the ranking of features can be concluded as: $\hat{d} > r_{\text{quality}} > t_{\text{rise}} > \xi_r > \kappa > \tau_{\text{RMS}} > r_{\max} > \tau_{\text{MED}}$. Notice that the proposed feature r_{quality} belongs to most of the best feature sets, indicating that it is highly informative for training regression models. Considering all feature set sizes, we select $\mathcal{F}_M^5$ as the best combination for regression and employ it in the subsequent experiments.

Model comparison. We compare the performance of SVM and MLP machine learning algorithms. For SVM

TABLE 5

Comparison of NLOS identification and mitigation performance using different models.

Model	NLOS-RMSE [m]	R^2	Accuracy [%]	training time [s]
Raw ranging error	5.27	–	–	–
SVM	1.07	0.97	98.57	59.94 (cls) 384.07 (reg)
MLP	1.05	0.97	99.84	879.40 (cls) 449.48 (reg)

TABLE 6

Comparison of NLOS identification and mitigation performance using different schemes.

Scheme	Iteration	NLOS-RMSE [m]	R^2	Accuracy [%]
Raw ranging error	–	5.27	–	–
Interp+Cls	0	1.17	0.96	97.43
	1	1.15	0.96	97.38
Interp+Map	0	1.17	0.96	97.43
	1	1.14	0.96	97.43
Prob+Cls	0	1.07	0.97	98.57
	1	1.05	0.97	97.55
Prob+Map	0	1.07	0.97	98.57
	1	1.06	0.97	97.71

model, the kernel function is selected as the radial basis function (RBF) and the penalty factor is set to 1. For MLP model, the activation transformation is realized by the sigmoid function and the learning procedure is optimized by the batch gradient descent algorithm. Table 5 shows the NLOS identification and mitigation performance using different models in *Test scene B*. We observe that both models can identify NLOS data accurately and reduce NLOS ranging errors effectively. In addition, the SVM model can achieve comparable performance to the MLP model in less time. Therefore, we employ the SVM model in the subsequent experiments.

Table 5 also describes the cost-effectiveness of our system. The cost of a learning-based localization system can be divided into two parts: labor cost and processing cost. In practice, the processing cost is far less than labor cost due to enhanced computing power. Existing solutions need to collect a large amount of labeled data for learning. In contrast, our system can automatically collect unlabeled data and produce weak labels through multi-sensory information fusion of PDR, UWB, and map data. We do not require labeled data, thereby greatly reducing the labor cost for database construction. The additional time of weak labels assignment is within a few hundred milliseconds. On the other hand, the processing cost mainly includes offline training cost and online positioning cost. As shown in Table 4, model training with tens of thousands of data can be completed in a few minutes. During the online information fusion positioning process, the computation time for each step is within tens of

TABLE 7
NLOS identification and mitigation performance of self-training.

Method	Iteration	Test scene A			Test scene B		
		NLOS-RMSE [m]	R^2	Accuracy [%]	NLOS-RMSE [m]	R^2	Accuracy [%]
Raw ranging error	–	0.91	–	–	5.27	–	–
Only PDR	0	1.08	0.76	92.60	1.07	0.97	98.57
PDR + UWB (with mitigation)	1	0.81	0.84	88.78	1.06	0.96	97.71
	2	0.71	0.86	89.15	1.06	0.96	98.28
	3	0.62	0.87	89.82	1.06	0.96	98.28
	4	0.59	0.87	90.48	1.06	0.97	98.28
	5	0.57	0.87	90.27	1.05	0.97	98.28
	6	0.56	0.87	90.32	1.05	0.97	98.28
	7	0.55	0.88	90.39	1.04	0.97	98.34

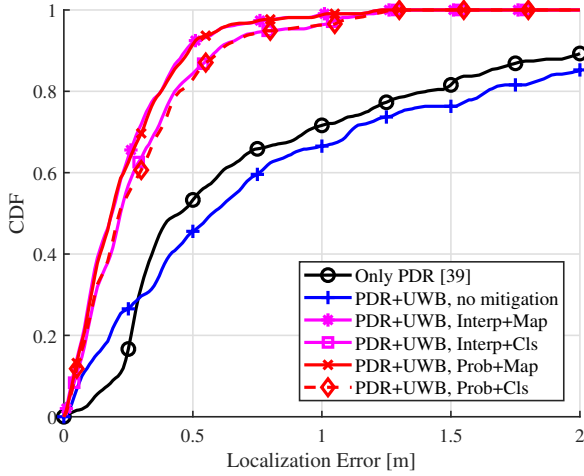


Fig. 6. Comparison of localization performance using different schemes. The black line represents the Map/PDR integrated system in [39]. The blue line represents the Map/PDR/UWB integrated system without NLOS mitigation. The other lines represent the Map/PDR/UWB integrated system with our NLOS mitigation method.

milliseconds.

Scheme comparison. We compare the performance of different labeling schemes and identification schemes. We consider four scheme combinations: (i) "Interp+Cls" uses interpolation labeling scheme and classifier-based identification scheme; (ii) "Interp+Map" uses interpolation labeling scheme and map-based identification scheme; (iii) "Prob+Cls" uses probability labeling scheme and classifier-based identification scheme; and (iv) "Prob+Map" uses probability labeling scheme and map-based identification scheme. Table 6 shows the NLOS identification and mitigation performance using different schemes in the first two iterations in *Test scene B*. By comparison, the probability labeling scheme exhibits better prediction performance than the interpolation labeling scheme. Furthermore, Fig. 6 shows the localization performance using different schemes in *Test scene B*. For a fair comparison, the raw UWB ranging estimates are fused into the system in the same way as the mitigation case. The result of only using PDR obtains the

positioning RMSE of 1.20m. Due to large ranging errors in indoor scenario, the result of directly fusing raw UWB ranging estimates without mitigation obtains the worse positioning RMSE of 1.29m. However, the overall localization accuracy is significantly improved after leveraging the proposed self-trained models. The four scheme combinations obtain the positioning RMSE of 0.31m, 0.38m, 0.31m and 0.39m in order of the legend. We observe that the map-based identification scheme exhibits better localization performance than the classifier-based identification scheme. Therefore, the optimal combination is probability labeling scheme aided with map-based identification scheme. We employ these schemes in the subsequent experiments.

4.4 Performance of models based on self-training

We calculate the prediction scores during eight iterations to assess the impact of our self-training strategy on model performance, as shown in Table 7. Notice that iteration 0 shows a higher RMSE of 1.08m in *Test scene A* since the baseline NLOS error is comparable to the positioning error when only using PDR. However, the RMSE decreases to a lower 0.81m after fusing the mitigated UWB measurements. Furthermore, the proposed self-training process is able to continuously reduce the RMSE to 0.55m, obtaining an improvement of 40% compared to raw ranging errors. For the more complex *Test scene B*, the RMSE shows a continuous decrease during self-training even with a narrow positioning area in a corridor, obtaining an improvement of 80% compared to raw measurement errors. Other metrics including R^2 and classification accuracy also show continuous improvement. These results show that the regression models can significantly reduce NLOS ranging errors even with incorrect labels, and the performance can be stably improved. The effectiveness is due to the automatic weak-labeling procedure and label quality improvement described in Section 4.3. Moreover, our results also show that the classification models can obtain satisfactory accuracy even with inaccurate labels because SVM classifiers can tolerate some wrongly labeled samples.

Fig. 7 shows the CDFs of the NLOS ranging error before and after mitigation. For *Test scene A*, the probability of errors smaller than 1m increases from 80% to 99%. For *Test*

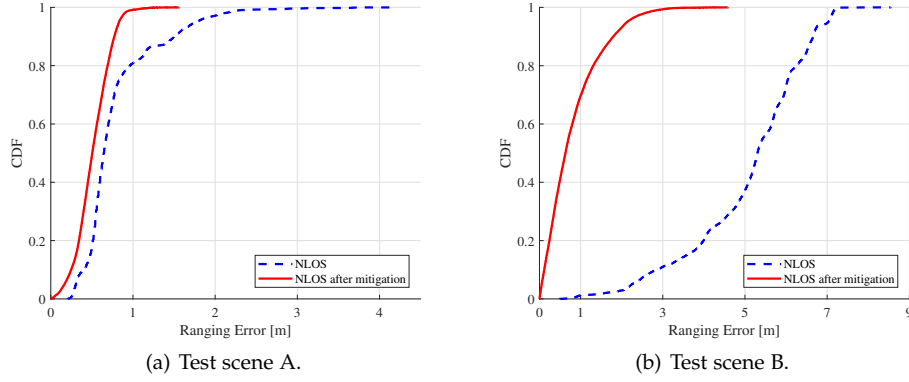


Fig. 7. CDFs of the NLOS ranging error before and after mitigation.

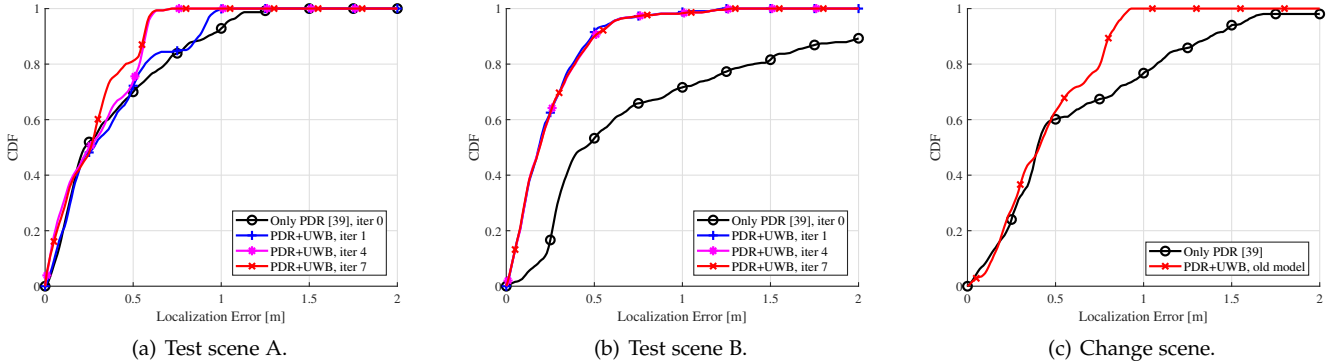


Fig. 8. Localization performance of self-training.

scene B, the probability of errors smaller than 1m increases from 1% to 70%. Therefore, the self-training strategy can significantly improve the NLOS mitigation.

4.5 Performance of localization based on self-training

We analyze the positioning error during eight iterations to assess the impact of our self-training strategy on localization performance, as shown in Fig. 8. The location probability distribution is initialized with weights on the four grid points surrounding the initial location. In the simple *Test scene A* (Fig. 8a), there are around 75% of the positioning results with an accuracy smaller than 0.6m when only using the PDR algorithm in [39]. Whereas, the proportion can be improved to 99% after several iterations of the proposed ranging mitigation and data fusion. In the complex *Test scene B* (Fig. 8b), the positioning results with an accuracy smaller than 0.5m can be improved from 53% to 90%. Fig. 9 shows the tracking results under multiple situations. The PDR algorithm suffers from long-term error accumulation, leading to room-level positioning errors as shown in Fig. 9e. When directly fusing UWB data without mitigation, NLOS effects may be alleviated by specific geographical constraints (Fig. 9b) or results in even worse localization results (Fig. 9f). In contrast, our method can automatically mitigate NLOS ranging errors and correct trajectories through self-training. After several iterations, high-accuracy tracking results are obtained in multiple different environments as shown in

Figs. 9d and 9h. These results show that the proposed techniques can significantly improve the localization accuracy in different real scenarios.

We further test the robustness and practicality of our system in additional and changing environments. Fig. 8c illustrates the localization performance in a changing environment. We collect 26 new trajectories with around 30000 UWB measurements in *Test scene A*. The testing scenarios are varied including: (i) changed positions of obstacles and NLOS propagation conditions; (ii) changed tester with different gender, height, and weight; (iii) changed smartphone from Google Nexus 6p to Xiaomi Mi 10 with Android 11; (iv) different positions of UWB tags such as wearing on the chest or carrying in the hand; (v) different locations with random trajectories; and (vi) multiple pedestrians moving freely in the room. In Fig. 8c, the red line represents the localization results when using the old model obtained in *Test scene A* (Fig. 8a) to mitigate UWB measurements in varied testing scenarios. We observe that even under such changes, the positioning results with errors smaller than 0.8m can be improved from 70% to 90%. The results demonstrate that our method is robust to changing environments without the need of data from new locations.

Fig. 10 presents additional experimental results in a totally new *Test scene C* which requires embedding of a new map. *Test scene C* is a hall with a dimension of $9.0\text{ m} \times 17.0\text{ m}$, as shown in Figs. 10a and 10b. In this new scenario, we collect 30 new trajectories and 34506 UWB measurements. The

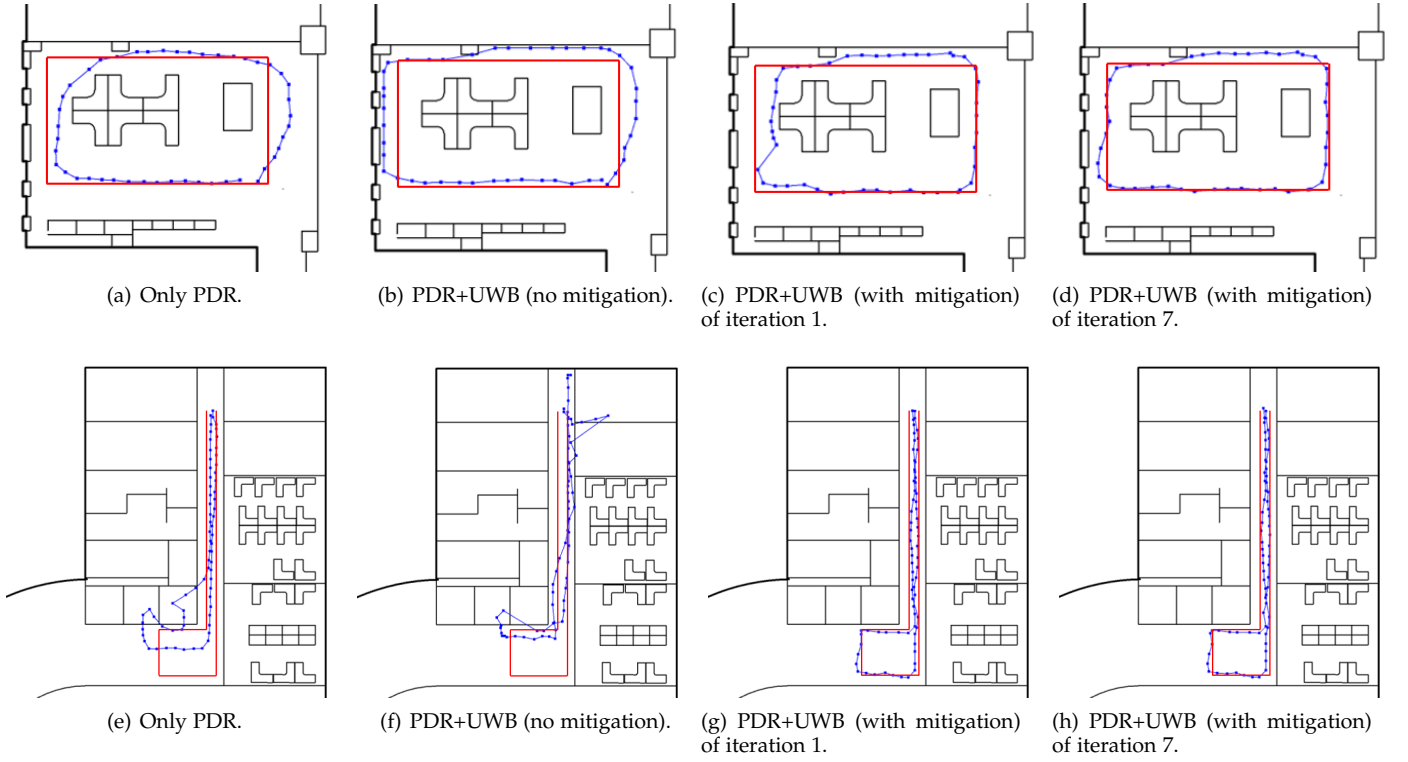


Fig. 9. Trajectories under different situations in Test scene A (upper) and Test scene B (lower). The blue lines represent the tracking results, and the red lines represent the ground truth.

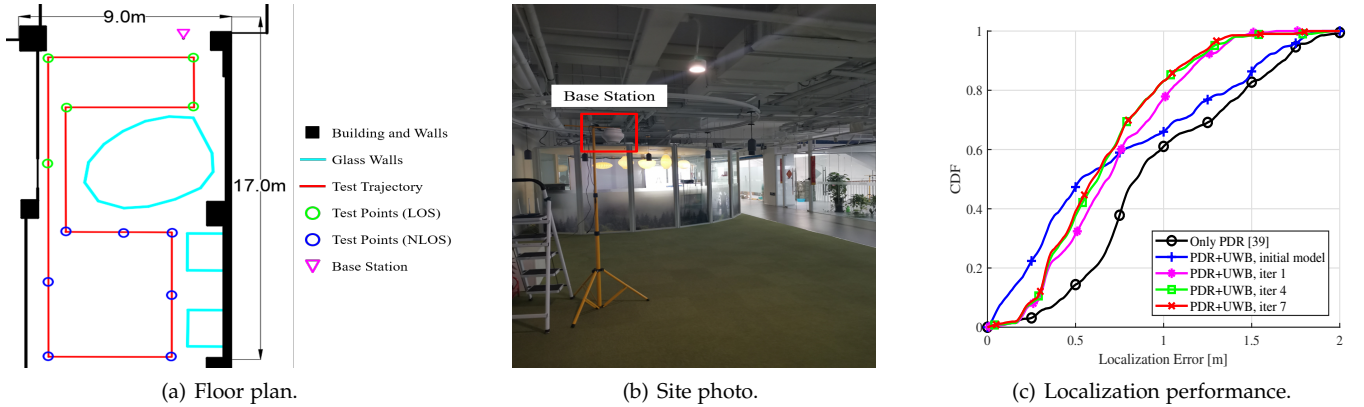


Fig. 10. Additional experimental site. Test scene C is a hall as shown in (a) and (b). The corresponding localization performance is shown in (c).

new testing scenario contains obstacles with different materials such as glass walls. The location of UWB base station is also changed to cover the new area. Fig. 10c illustrates the localization performance in *Test scene C*, where “PDR+UWB, initial model” describes the performance achieved using the model obtained in *Test scene A* (Fig. 8a) to mitigate UWB measurements in *Test scene C*. The results show that even in a totally new environment, the proposed techniques can successfully be initialized using a model obtained in other environment. In particular, with the self-training strategy, the proposed techniques can quickly adapt to the new environment and obtain higher localization accuracy. As shown in the figure, after some iterations the proposed techniques significantly reduce the outage probability of errors larger than 1m, which demonstrates that our method

can automatically adapt to a new practical environment.

We also compare our self-training method with traditional supervised learning methods. In particular, we created a new database in which an accurate label is assigned to each sample based on the knowledge of the trajectory. All trajectory data of each scene are randomly divided into training set and test set at a ratio of around 80:20. Specifically, for *Test scene A*, the training set consists of around 40000 accurately or weakly labeled data obtained from 15 trajectories and the test set consists of around 10000 UWB data corresponding to 11 trajectories. For *Test scene B*, the training set consists of around 16000 accurately or weakly labeled data obtained from 12 trajectories and the test set consists of around 4000 UWB data corresponding to 9 trajectories. For *Test scene C*, the training set consists

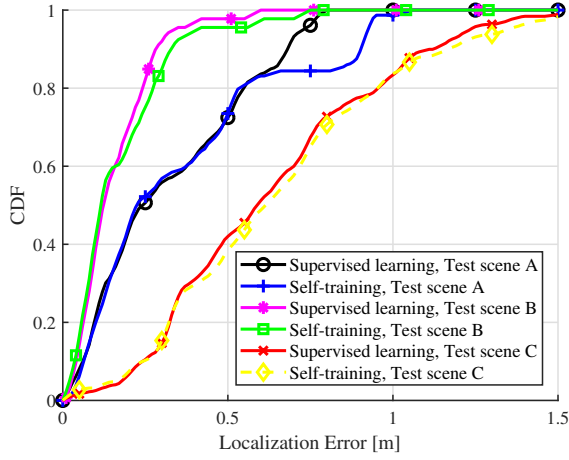


Fig. 11. Comparison with supervised learning.

of around 60000 accurately or weakly labeled data obtained from 16 trajectories and the test set consists of around 15000 UWB data corresponding to 14 trajectories. We use the same testing data to compare the proposed self-training method that learns from weak labels with traditional supervised learning method that learns from accurate labels. Fig. 11 shows that the localization error of the proposed techniques is very close to that of supervised learning. These results validate the effectiveness of the proposed techniques in scenarios where accurately labeled data is not available.

4.6 Generalization to online learning

The above experiments perform self-training on the whole collected training data, which is referred as a batch mode. In common practical situations, there is not enough data when facing a new environment. The proposed multi-sensory information fusion system has the capability to deal with these cases. Trajectories can be sequentially fed into the system and are expected to help train the model from the perspective of online learning. When the first user enters a new environment without any prior knowledge on NLOS mitigation, the system can still work by using map and inertial sensors. During the positioning process, the system automatically collects unlabeled UWB data and produces weak labels to train the mitigation model. In the next iteration, the mitigated UWB measurements can be integrated into the system for information fusion and weak labels refinement. This process is performed iteratively based on the self-training strategy.

We design an online learning process for self-training as follows: (i) initialize UWB models with the PDR tracking results of the first user in iteration 0; (ii) use existing models to carry out ranging error mitigation and data fusion positioning for the current user; (iii) correct the historical trajectories of the former users; (iv) update models based on the data from step (ii) and (iii); and (v) go back to step 2) for the next user. We use the data of *Test scene B* that contains multiple users passing through the same area for evaluation. Fig. 12 illustrates the changes of model performance during online learning. We add one new trajectory each time, thus the total number of iterations is equal to that of trajectories.

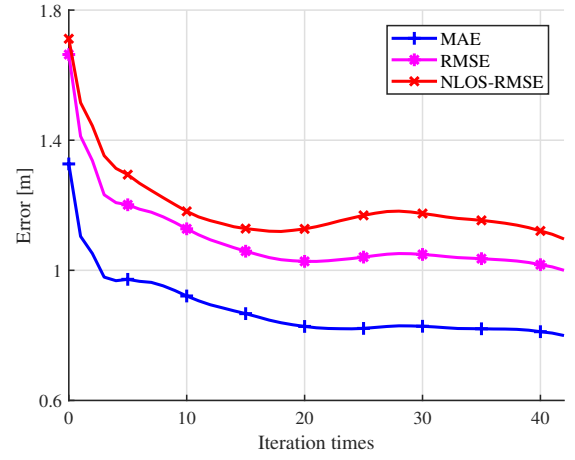


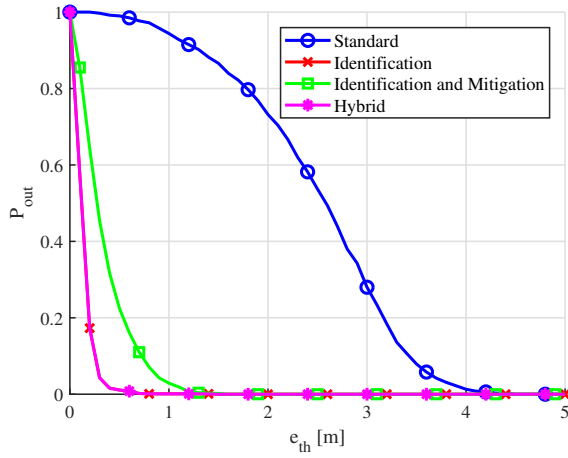
Fig. 12. Changes of model performance during online learning.

Notice that the amount of training data increases linearly with the number of iterations. As expected, the prediction errors on the test data set shows a downward trend. The effectiveness of this approach is due to the increase in size and quality of training data. The RMSE of NLOS ranging errors can be quickly decreased from 1.71m to 1.18m, obtaining an improvement of 31% after 10 trajectories. Results demonstrate that our method can quickly provide models adapted to the new environment even without any prior information. Some noisy trajectories may result in bad localization results to damage model performance, such as the fluctuation around the iteration 25. However, this short-term error can be automatically corrected after a period of time, demonstrating the robustness of our method.

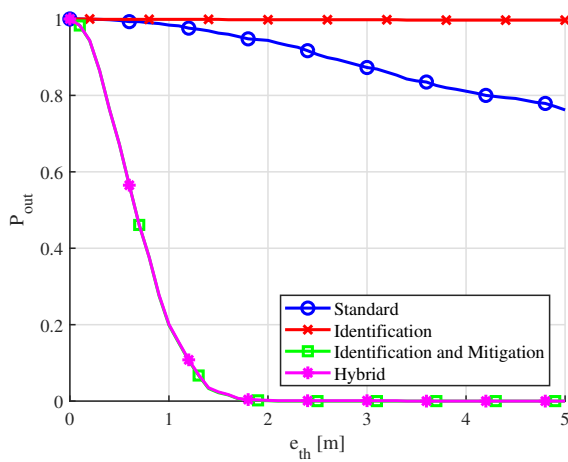
4.7 Generalization to multi-base station localization

In a multi-base station scenario, user location can be inferred based on multiple distance estimates. We follow the simulation setup in [18], [25] to evaluate the effectiveness of our method on multi-base station localization. We locate the user at $\mathbf{p} = (0, 0)$, and place N_b base stations around the user with equal angular spacing. For each base station, we draw a waveform with probability P_{NLOS} from the NLOS database, and with probability $1 - P_{\text{NLOS}}$ from the LOS database. The true distance d_i corresponding to that waveform is then used to place the i -th base station at position $\mathbf{p}_i = (d_i \cos(2\pi(i-1)/N_b), d_i \sin(2\pi(i-1)/N_b))$, while the ranging estimate $\hat{d}_i$ is provided to the user for localization. The localization problem can be solved based on Least Squares (LS) algorithm and initialized with the arithmetic mean of all base station positions.

We compare the performance of four localization strategies: (i) "Standard", a baseline method that uses all the ranging estimates for localization; (ii) "Identification", a method that uses only the ranging estimates classified as LOS by our identification model for localization; (iii) "Identification and Mitigation", a method that uses the raw ranging estimates identified as LOS and the mitigated ones identified as NLOS for localization; and (iv) "Hybrid", a method that only uses the mitigated ranging estimates when less than three LOS base stations are available, combining (ii) and (iii). Notice



(a) Outage probability for $N_b = 5$ anchors, with $P_{\text{NLOS}} = 0.2$.



(b) Outage probability for $N_b = 5$ anchors, with $P_{\text{NLOS}} = 0.8$.

Fig. 13. Comparison of different multi-base station localization strategies varying NLOS probability.

that in case (ii), the available distance estimates may be less than three due to directly discarding the data identified as NLOS. The system is unable to locate and we set the localization error to $+\infty$.

Fig. 13 shows the comparison results of different localization strategies varying the NLOS probability on the database of *Test scene B*. We utilize outage probability [18], [25] to assess the accuracy and availability of localization. This metric represents the expected number of positions with error exceeding e_{th} , given by $P_{\text{out}}(e_{\text{th}}) = \mathbb{E}\{\mathbb{1}_{\{\|\mathbf{p}-\hat{\mathbf{p}}\| > e_{\text{th}}\}}\}$ where $\mathbb{1}$ is the indicator function. The results of the "Standard" strategy indicate that the existence of NLOS propagation causes performance degradation. When most base stations are blocked as shown in Fig. 13b, the "Identification" strategy fails to locate due to insufficient LOS measurements. However, after using the models obtained from our system, the "Identification and Mitigation" strategy significantly improves the localization performance. The results show that in 80% of the cases the localization accuracy is within 1m even with a high NLOS probability ($P_{\text{NLOS}} = 0.8$). Considering the fact that ranging error can not be completely eliminated, the "Hybrid" strat-

egy leads to better performance with low NLOS probability as shown in Fig. 13a.

5 CONCLUSION

This paper proposed a novel indoor localization system with NLOS mitigation based on self-training. The system combines information from map, inertial sensors, and UWB transceivers to infer pedestrian trajectories. The proposed method enables automatic data collection, labeling, and learning through multi-sensory information fusion. The weak labels for UWB measurements are produced and iteratively refined by self-training, thereby significantly reducing the labor cost of traditional supervised learning methods. Experimental results demonstrated that our method can significantly reduce NLOS ranging error, improve localization accuracy, and enable multi-base station localization under mixed LOS/NLOS conditions. Our results show that self-training methods can enable indoor localization and tracking with high-accuracy, low-cost, and strong-robustness.

The future work may include three parts. Firstly, the experimental scenes can be enriched and large amounts of data can be collected by crowdsourcing to develop the method for large-scale problems. Secondly, more complex machine learning algorithms such as gradient boosting decision trees (GBDT), convolutional neural networks (CNN), and minimax risk classifiers (MRC) can be explored to improve the feature representation and model performance. Thirdly, other RF signals such as 5G, Wi-Fi, and Bluetooth as well as other signal metrics such as strength, phase, and fingerprint can be fused to further improve the proposed localization system.

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REFERENCES

- [1] M. Z. Win, Y. Shen, and W. Dai, "A theoretical foundation of network localization and navigation," *Proc. IEEE*, vol. 106, no. 7, pp. 1136–1165, Jul. 2018.
- [2] Q. Xiao, B. Xiao, J. Cao, and J. Wang, "Multihop range-free localization in anisotropic wireless sensor networks: A pattern-driven scheme," *IEEE Trans. Mobile Comput.*, vol. 9, no. 11, pp. 1592–1607, Nov. 2010.
- [3] S. Gezici *et al.*, "Localization via ultra-wideband radios: A look at positioning aspects for future sensor networks," *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 70–84, Jul. 2005.
- [4] A. Conti, S. Mazuelas, S. Bartoletti, W. C. Lindsey, and M. Z. Win, "Soft information for localization-of-things," *Proc. IEEE*, vol. 107, no. 11, pp. 2240–2264, Sep. 2019.
- [5] Y. Shen and M. Z. Win, "Fundamental limits of wideband localization – Part I: A general framework," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4956–4980, Oct. 2010.
- [6] C. Laoudias *et al.*, "A survey of enabling technologies for network localization, tracking, and navigation," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3607–3644, Jul. 2018.
- [7] H. Chenji and R. Stoleru, "Toward accurate mobile sensor network localization in noisy environments," *IEEE Trans. Mobile Comput.*, vol. 12, no. 6, pp. 1094–1106, Jun. 2013.

- [8] F. Shahzad, T. R. Sheltami, and E. M. Shakshuki, "Dv-maxhop: A fast and accurate range-free localization algorithm for anisotropic wireless networks," *IEEE Trans. Mobile Comput.*, vol. 16, no. 9, pp. 2494–2505, Sep. 2017.
- [9] C. Yang and H. Shao, "WiFi-based indoor positioning," *IEEE Commun. Mag.*, vol. 53, no. 3, pp. 150–157, Mar. 2015.
- [10] C. Zhang, M. J. Kuhn, B. C. Merkl, A. E. Fathy, and M. R. Mahfouz, "Real-time noncoherent UWB positioning radar with millimeter range accuracy: Theory and experiment," *IEEE Trans. Microw. Theory Techn.*, vol. 58, no. 1, pp. 9–20, Jan. 2010.
- [11] S. S. Saab and Z. S. Nakad, "A standalone RFID indoor positioning system using passive tags," *IEEE Trans. Ind. Electron.*, vol. 58, no. 5, pp. 1961–1970, May 2011.
- [12] R. Zhang, F. Hoflinger, and L. Reindl, "Inertial sensor based indoor localization and monitoring system for emergency responders," *IEEE Sensors J.*, vol. 13, no. 2, pp. 838–848, Feb. 2013.
- [13] J. Dong, M. Noreikis, Y. Xiao, and A. Ylä-Jääski, "Vinav: A vision-based indoor navigation system for smartphones," *IEEE Trans. Mobile Comput.*, vol. 18, no. 6, pp. 1461–1475, Jun. 2019.
- [14] S. J. Kim and B. K. Kim, "Dynamic ultrasonic hybrid localization system for indoor mobile Robots," *IEEE Trans. Ind. Electron.*, vol. 60, no. 10, pp. 4562–4573, Oct. 2013.
- [15] Y. Wu, Q. Chaudhari, and E. Serpedin, "Clock synchronization of wireless sensor networks," *IEEE Signal Process. Mag.*, vol. 28, no. 1, pp. 124–138, Jan. 2011.
- [16] I. W. Group *et al.*, "IEEE standard for local and metropolitan area networks—Part 15.4: Low-rate wireless personal area networks (lr-wpans)," *IEEE Std 802.15.4-2011 (Revision of IEEE Std 802.15.4-2006)*, pp. 1–314, Sep. 2011.
- [17] A. Yassin *et al.*, "Recent advances in indoor localization: A survey on theoretical approaches and applications," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 24, pp. 1327–1346, Secondquarter 2017.
- [18] S. Marano, W. M. Gifford, H. Wymeersch, and M. Z. Win, "Nlos identification and mitigation for localization based on UWB experimental data," *IEEE J. Sel. Areas Commun.*, vol. 28, no. 7, pp. 1026–1035, Sep. 2010.
- [19] S. Wu, Y. Ma, Q. Zhang, and N. Zhang, "Nlos error mitigation for UWB ranging in dense multipath environments," in *Proc. IEEE Wireless Commun. and Networking Conf.*, Kowloon, China, Mar. 2007, pp. 1565–1570.
- [20] J. Schroeder, S. Galler, K. Kyamakya, and K. Jobmann, "Nlos detection algorithms for ultra-wideband localization," in *Proc. Workshop Pos., Nav. Commun.*, Hannover, Germany, Mar. 2007, pp. 159–166.
- [21] J. Khodjaev, Y. Park, and A. S. Malik, "Survey of NLOS identification and error mitigation problems in UWB-based positioning algorithms for dense environments," *Annals of Telecommunications*, vol. 65, no. 5–6, pp. 301–311, Aug. 2009.
- [22] S. Venkatesh and R. M. Buehrer, "NLOS mitigation using linear programming in ultrawideband location-aware networks," *IEEE Trans. Veh. Technol.*, vol. 56, no. 5, pp. 3182–3198, Sep. 2007.
- [23] H. Wymeersch, S. Marano, W. M. Gifford, and M. Z. Win, "A machine learning approach to ranging error mitigation for UWB localization," *IEEE Trans. Commun.*, vol. 60, no. 6, pp. 1719–1728, Jun. 2012.
- [24] C. Mao, K. Lin, T. Yu, and Y. Shen, "A probabilistic learning approach to UWB ranging error mitigation," in *Proc. IEEE Global Commun. Conf.*, Abu Dhabi, United Arab Emirates, Dec. 2018, pp. 1–6.
- [25] S. Mazuelas, A. Conti, J. C. Allen, and M. Z. Win, "Soft range information for network localization," *IEEE Trans. Signal Process.*, vol. 66, no. 12, pp. 3155–3168, Jun. 2018.
- [26] Z. H. Zhou, "A brief introduction to weakly supervised learning," *National Science Review*, vol. 5, no. 1, pp. 44–53, Jan. 2018.
- [27] S. Mazuelas and A. Pérez, "General supervision via probabilistic transformations," in *24th European Conference on Artificial Intelligence-ECAI 2020*, Santiago de Compostela, Spain, Aug. 2020, pp. 1348–1354.
- [28] R. T. Sousa and J. Gama, "Comparison between co-training and self-training for single-target regression in data streams using AMRules," in *IOTSTREAMING@PKDD/ECML*, 2017.
- [29] Y. Gu, Y. Chen, J. Liu, and X. Jiang, "Semi-supervised deep extreme learning machine for Wi-Fi based localization," *Neurocomputing*, vol. 166, pp. 282–293, Oct. 2015.
- [30] M. Mohammadi, A. Al-Fuqaha, M. Guizani, and J. Oh, "Semisupervised deep reinforcement learning in support of IoT and smart city services," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 624–635, Apr. 2017.
- [31] J. J. Pan, S. J. Pan, J. Yin, L. M. Ni, and Q. Yang, "Tracking mobile users in wireless networks via semi-supervised colocalization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 3, pp. 587–600, Mar. 2011.
- [32] J. Yoo and H. J. Kim, "Target localization in wireless sensor networks using online semi-supervised support vector regression," *IEEE Sensors J.*, vol. 15, no. 6, pp. 12 539–12 559, May 2015.
- [33] X. Ye, S. Huang, Y. Wang, W. Chen, and D. Li, "Unsupervised localization by learning transition model," *Proc. ACM Interactive Mobile Wearable Ubiquitous Tec.*, vol. 3, no. 2, pp. 1–23, Jun. 2019.
- [34] T. Wang, K. Hu, Z. Li, K. Lin, J. Wang, and Y. Shen, "A semi-supervised learning approach for UWB ranging error mitigation," *IEEE Wireless Commun. Lett.*, vol. 10, no. 3, pp. 688–691, Mar. 2020.
- [35] S. Mazuelas, Y. Shen, and M. Z. Win, "Belief condensation filtering," *IEEE Trans. Signal Process.*, vol. 61, no. 18, pp. 4403–4415, Sep. 2013.
- [36] B. Ristic, S. Arulampalam, and N. Gordon, *Beyond the Kalman filter: Particle filters for tracking applications*. Norwood, MA, USA: Artech house, 2003.
- [37] M. Z. Win *et al.*, "Network localization and navigation via cooperation," *IEEE Commun. Mag.*, vol. 49, no. 5, pp. 56–62, May 2011.
- [38] F. Ge and Y. Shen, "Single-anchor ultra-wideband localization system using wrapped PDoA," *IEEE Trans. Mobile Comput.*, 2021, Early Access.
- [39] —, "Real-time indoor localization on smartphones by multi-grained grid-based filters," in *Proc. IEEE Global Commun. Conf.*, Abu Dhabi, United Arab Emirates, Dec. 2018, pp. 1–6.
- [40] R. W. Levi and T. Judd, "Dead reckoning navigational system using accelerometer to measure foot impacts," Dec. 10 1996, uS Patent 5,583,776.
- [41] W. Kang and Y. Han, "Smartpdr: Smartphone-based pedestrian dead reckoning for indoor localization," *IEEE Sensors J.*, vol. 15, no. 5, pp. 2906–2916, May 2015.
- [42] W. Zijlstra, "Assessment of spatio-temporal parameters during unconstrained walking," *Eur. J. Appl. Physiol.*, vol. 92, no. 1–2, pp. 39–44, Jun. 2004.
- [43] V. Kubelka and M. Reinstein, "Complementary filtering approach to orientation estimation using inertial sensors only," in *Proc. IEEE Int. Conf. Robot. and Autom.*, Saint Paul, MN, USA, May 2012, pp. 599–605.
- [44] D. Neiryneck, E. Luk, and M. McLaughlin, "An alternative double-sided two-way ranging method," in *Proc. Workshop Pos., Nav. Commun.*, Bremen, Germany, Oct. 2016, pp. 1–4.
- [45] F. Liu, J. Wang, J. Zhang, and H. Han, "An indoor localization method for pedestrians base on combined UWB/PDR/Floor map," *Sensors*, vol. 19, no. 11, p. 2578, Jun. 2019.
- [46] H. Tong, N. Xin, X. Su, T. Chen, and J. Wu, "A robust pdr/uwb integrated indoor localization approach for pedestrians in harsh environments," *Sensors*, vol. 20, no. 1, p. 193, Dec. 2020.
- [47] C. Wang *et al.*, "A high-accuracy indoor localization system and applications based on tightly coupled UWB/INS/Floor map integration," *IEEE Sensors J.*, vol. 21, no. 16, pp. 18 166–18 177, May 2021.
- [48] DW1000 User Manual (2.11), Decawave Ltd, Dublin, Ireland, 2017.
- [49] N. Cristianini and J. Shawe-Taylor, *An introduction to support vector machines and other kernel-based learning methods*. Cambridge, UK: Cambridge university press, 2000.
- [50] V. Vapnik, *The nature of statistical learning theory*. Berlin/Heidelberg, GER: Springer science & business media, 2013.
- [51] S. Mazuelas, Y. Shen, and A. Pérez, "Generalized maximum entropy for supervised classification," *IEEE Trans. Inf. Theory*, 2021.
- [52] C. M. Bishop, *Pattern recognition and machine learning*. New York, NY, USA: springer, 2006.



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